

Data Profiling and Key Analysis of MBA Acceptance

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ABSTRACT

As the field of Artificial Intelligence shifts from a model-centric to a data-centric approach, assessing and improving data quality becomes a critical component to enhance the performance of AI systems. This report examines the MBA Acceptance dataset, identifying several key data quality issues that may negatively impact the accuracy and robustness of trained models. Through a systematic data assessment, issues such as data imbalance, underrepresented subgroups, and multicollinearity were detected. This report hypothesises what possible problems might arise from these data problems, and what strategies should be employed to mitigate these issues and improve the overall quality of the dataset.

1 INTRODUCTION

Nowadays we are moving from a Model-Centric to a Data-Centric Artificial Intelligence[3], and with that comes the need to evaluate and assess the quality of the data utilized. It is ever more apparent that an AI Model is only as good as the quality of the data used to train it, and that the biggest potential on improvement for AI is achieved by improving its data quality.

With that in mind, it was chosen a dataset with potential to do a deep analysis on potential data issues it might have, and what technical and social problems it could cause. The chosen dataset was "MBA Admission, Class 2025"[4], which corresponds to synthetic data generated from the Wharton Class of 2025's statistics.

As it is stated in the title of the dataset, it comprises of data from admission to a MBA program, being this data part of the theme on *Education Access*. If there are issues in these data, this could pose a problem to algorithms trained on this dataset, since they could contain biases in certain variables that might put at risk the fairness of the algorithm, and contribute to perpetuating societal inequalities.

Therefore, it is important to do a correct data profiling of the dataset, in order to have a clearer view of the problems and qualities which exist inside the data.

2 DATA PROFILING SUMMARY

The data profiling of the dataset was done using two main methods: one simple, utilizing pandas and describing the data on a dataframe with other small analysis; and one more advanced, taking advantage of the `ydata_profiling`[1] package to do more extensive exploration (like multivariate analysis) in a friendlier UI.

This dataset consists of **6194 candidates** and their social, ethnic, career and academic entries, with the respective result of their application (admitted, wait-listed or rejected). More specifically, the following table shows the metadata features present in the dataset in study.

Table 1: Metadata Description

Metadata Column	Description	Constraints
application_id	application ID	UNIQUE
gender	Applicant's gender	Male, Female
international	International student	TRUE/FALSE
gpa	Grade Point Average	4.0 scale
major	Undergraduate major	-
race	Racial background	null: international
gmat	GMAT score	800 points scale
work_exp	Work Experience	Years
work_industry	Previous Industry	-
admission	Admission status	null: Rejected

As it is possible to observe in the table above, some features have empty values not because of poor data quality, but because it was conveyed that the null value would have that representation. That is the case for the race and admission features. In addition, there are no duplicated rows in the dataset, which contributes to the overall perceived quality of it.

Starting with the categorical variables, in the gender field, we can observe that about 63.7% of the dataset is comprised of men, while the remaining 36.3% are women. In regard to internationality, about 70.3% are locals, while 29.7% are international applicants. Moreover, in the category of major, 40.1% of applicants are from Humanities, 30.3% graduated from STEM areas, and 29.7% from Business. About 23.5% of applicants were White, 18.5% were Asian, 14.8% were Black, 9.6% were Hispanic and 3.8% were from other ethnical background. The remaining 29.7% correspond to the number of international students (with no data on their ethnicity). Finally, only 900 applicants (14.5%) were accepted to do the MBA, 100 applicants (1.6%) were wait-listed and the remaining 83.9% were rejected.

Then it is time to analyse the quantitative variables. To sum up, the average GPA is 3.25, the average GMAT score is 651, the average work experience is 5.02 years. All curves appear to have a normal distribution, except the GMAT curve, which appears to have more results (compared to their neighbours) in the lowest end of the spectrum. This might indicate some results were skewed to be at a certain threshold (either a passing grade, or a minimum requirement for the MBA).

3 KEY ISSUES AND ANALYSIS

This dataset will be put to test against tree possible data issues: Data Imbalance, Underrepresented Subgroups of Data and Multicollinearity. There are other issues in this dataset, such as the case of missing data. As it was discussed previously, some entries are

empty to represent some information (for example, rejections are represented as an empty cell in the admissions column). This may cause problems to models running on this dataset (if they don't interpret correctly this information), or in the worst case, there might be ambiguity between a row simply lacking data and in this case it wouldn't be seen a mistake (for example, if a student is missing its race cell, it would be considered as an international student). However, since there is also a column to distinguish if a student is international or not, this problem would be mitigated. Nonetheless, this unnecessary column will create a problem of Multicollinearity, further analysed in 3.3.

3.1 Data Imbalance

This issue is the most evident in the dataset. For example, as analysed in section 2, the ratio of male applicants versus female applicants is almost 2:1, and the data imbalance is even more evident in the target class (admission). Approximately 14.5% of applicants are accepted while 85.5% are rejected or waitlisted, which causes a big imbalance. This culminates in the skewed features being more sensitive in the predictions, which makes it difficult for the model to predict accurately without creating some biases towards the majority class, achieving high accuracy while failing to effectively identify good candidate from minority classes.

Since this imbalance might be a real world representation, this issue can be fixed, for example, by oversampling the minority class. Despite not being viable to admit more people to the MBA programme, it could be a solution to try to incentivise the minorities present in the dataset (in this case, women) to apply for the MBA, such that it increases the number of applicants towards a more equal distribution. The same thing can be said for other features who are not completely balanced (such as race).

3.2 Underrepresented Subgroups of Data

A key issue in the dataset is the underrepresentation of certain subgroups within the admission class. After analysis of the distribution of admitted applicants based on gender and race, it becomes clear that some subgroups are not well-represented. This lack of representation can affect the fairness and accuracy of any predictive models developed using this data.

In this case, both White and Asian applicants have relatively equal numbers of admitted males and females, with 122 White males and 122 White females admitted, and 95 Asian males and 95 Asian females admitted. Despite this, as we move to other racial groups, the imbalance becomes more pronounced. Black and Hispanic applicants have much lower numbers of admitted individuals, with just 40 Black males and 40 Black females, and even fewer for Hispanic applicants: 31 males and 31 females. The "Other" racial group shows even bigger under-representation, with only 23 males and 23 females admitted.

As the previous subsection, this can cause problems in the accuracy of the prediction of models trained on this dataset. The subclasses with more data (in this case, White and Asian people) will have a favourable bias, which perpetuates societal inequalities against Hispanic and black people.

To solve these problems, we can employ a plurality of tactics, such as oversampling (or data augmentation), using techniques

like SMOTE (Synthetic Minority Over-sampling Technique)[2], applying class weights in the model to give more importance to the underrepresented subgroups, and therefore reducing the potential for bias in the predictive outcomes, or, as presented in subsection 3.1, trying to create initiatives to attract applicants from minority ethnical backgrounds, and having the overall ethnicity distribution in the MBA in consideration.

3.3 Multicollinearity

Multicollinearity happens when two (or more) independent variables in a dataset are highly correlated, leading to instability in predictive models. In this dataset, an example of this can be seen between the variables `international` and `race`, since they share a correlation of 100%. This happens, because the null values in the column `race` corresponds to the international students, as explained in the beginning of section 3.

Multicollinearity can distort the interpretation of model coefficients, which means it becomes harder to estimate the true impact of each variable on the target outcome. For instance, if a linear regression model were applied to this dataset, the coefficients for both `international` and `race` might be wrongfully estimated, leading to skewed results.

One way to address multicollinearity is to remove one of the correlated features, or to change the features in such a way that reduces correlation between them. In this case, the most obvious change to do is to remove the column `international` and change every null value from the column `race` to "International". Doing this, the information is kept the same, and the multicollinearity is removed, ensuring the model remains stable and interpretable.

4 CONCLUSIONS AND RECOMMENDATIONS

To conclude, this report was made with the purpose to identify key data quality issues in the MBA Acceptance dataset, and how they could be solved. The referred problems can make the predictions of AI systems unstable, and the results biased towards majority groups, perpetuating societal prejudices against minorities.

By employing tactics of oversampling (such as with the SMOTE technique), tuning model parameters to favour under-represented data, promoting data collection in the real world from minority groups in the dataset, and solving variable multicollinearity, we create better data and AI systems become fairer.

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